

Discovering Unusual Customer Purchasing Behavior



PROJECT PROPOSITION

Data Mining Project

CSCA 5522

Simon S.

sash1081@colorado.edu

University of Colorado, Boulder

Abstract

- Investigates unusual customer purchasing behavior in UK online retail transactions
- Invoice-level dataset: each transaction's own features + aggregated customer-level behavior
- Goal: discover/characterize behavioral segments, then find transactions & customers that don't fit any of them
- Not a predictive-modelling exercise; clustering is a means, not the end
- Main axes of interest: cancellation behavior and order total

Motivation: Business Value

- Identify high-value customers
- Identify segments associated with frequent cancellations
- Detect emerging changes in purchasing patterns
- Supports retention, marketing, and operational decisions

The Dataset

- `online_retail_II.csv` (Kaggle) - transaction log, Dec 2009 - Dec 2011
- ~1.07M line-item rows, 43 countries
- UK-only: 94.7% of all transactions; other countries too sparse for safe inference
- ~1/4 of UK rows are guest purchases (no Customer ID) - dropped
- After filtering: ~741,000 transaction rows -> ~40,500 invoices from ~5,400 customers

Unit of Analysis & Scope

- Unit of analysis: the individual invoice
- Described by its own transaction-level features + its customer's aggregated behavior
- A customer's routine invoices can cluster normally while a few of their invoices stand out as outliers
- Out of scope: regression on cancellation probability, basket prediction, clustering as an end in itself

Objective

- Identify and characterize purchasing behavior that differs substantially from the major segments
- Trace those outliers back to the customers who produced them
- Example: if 3 well-defined clusters emerge - which transactions/customers fit none of them?

Related Work

- [Customer Segmentation & Retention Strategy](#) (Kaggle) - segments as end product for retention
- [Online Retail Cohort Analysis](#) (Kaggle, same dataset) - descriptive cohorts/trends, not segmentation+outliers
- [Enhanced RFM / RFMT](#) (Kaggle) - adds Inter-Purchase-Time to RFM; segmentation is the end goal there
- [Recommendation Systems](#) (Kaggle) - association rules & collaborative filtering, different question entirely
- This project differs: segmentation is a baseline, not the deliverable - the deliverable is what doesn't fit

Proposed Work: Feature Engineering

- Filter to UK, drop guest transactions
- Build customer-level features:
 - Timing: day of week, hour, avg. shopping trips per month
 - Spending: avg/min/max/std of transaction totals
 - Product: avg/min/max/std of items purchased, most frequent products
 - Cancellations: % of cancelled transactions
- Join customer-level features back onto invoice-level records
- Feasibility: ~50 engineered features, ~40,500 invoices, ~5,400 customers

Proposed Work: Analysis Flow

- Explore segmentation using engineered behavioral features
- Characterize the common patterns found in each segment
- Identify transactions/customers whose behavior differs significantly from the majority or from any segment
- Particular focus: spending, purchase composition, cancellation behavior

Evaluation: Clustering

- Baseline: KMeans (quick, cheap, first impression - cannot produce outliers)
- Working model: DBSCAN (produces outliers - the “working horse” of this research)
- Two distance metrics evaluated: euclidean and cosine
 - DBSCAN supports cosine natively; “cosine KMeans” approximated via L2-normalization + euclidean KMeans
- Evaluated on: silhouette score (direction, not verdict), stability under hyper-parameter changes, interpretability
- Target: stable, 2-3 interpretable clusters, manageable/interpretable outliers

Evaluation: Characterizing Differences

- Question: how does cluster A differ from cluster B? How do outliers differ from inliers?
- Raw mean differences aren't comparable across features on different scales
- Solution: Cohen's d - standardized mean difference, comparable across features
- Take the top 5-10 features with $|\text{Cohen's } d| > 0.8$ to characterize/define each population

Discussion & Conclusion

- To be completed once clustering, outlier detection, and Cohen's d analyses are finalized

References

- Full reference list in the project proposition document (`project_proposition_draft.md`, Section 7)
- Covers: PCA (Jolliffe 2002), KMeans (MacQueen 1967), DBSCAN (Ester et al. 1996; Schubert et al. 2017), silhouette score & DBSCAN docs (scikit-learn), Cohen's d (Cohen 1988), explainable anomaly detection (Li et al. 2023)